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***** Cloud Engineer Interview Questions *****

Important Note:

You will find AWS services naming convention here. If you are preparing for Azure or GCP, you can still use these questions—just change the names to match your cloud provider.

All the very best – Keep Hustling and don't lose hope

LEVEL 1 Easy - Beginner Level Questions

(Concept clarity level)

1. What is cloud computing?
2. What are IaaS, PaaS, and SaaS?
3. What is a Region and Availability Zone?
4. What is a VPC?
5. Difference between public and private subnet?
6. What is an Internet Gateway?
7. What is EC2?
8. What is S3?
9. Difference between S3, EBS, and EFS?
10. What is IAM?
11. What is a Security Group?
12. What is Auto Scaling?
13. What is a Load Balancer?
14. What is CIDR?
15. What is high availability?
16. Why do companies move from on-premises to cloud?
17. How would you explain cloud computing to a non-technical person?
18. Why do we need multiple Availability Zones?
19. Why is a VPC required before launching resources?
20. When would you use a public subnet vs a private subnet?
21. Why can't a private subnet access the internet directly?
22. Why is an Internet Gateway attached to a VPC and not to a subnet?
23. Why do we need route tables in networking?
24. How does a Security Group protect an EC2 instance?
25. Why are Security Groups called stateful?
26. Why is IAM role safer than sharing access keys?
27. When would you choose S3 over EBS?
28. Why do we need Auto Scaling even if one server is working fine?

29. How does a Load Balancer improve availability?
30. Why is high availability important for production applications?
31. How does horizontal scaling differ in impact compared to vertical scaling?
32. Why is monitoring necessary in cloud environments?
33. How would you secure a basic web server in the cloud?
34. Why should secrets not be hardcoded inside applications?
35. How does cloud help reduce downtime compared to traditional servers?
36. Difference between NACL and Security Group
37. Difference between CloudFormation and Terraform
39. Could you please list down 6 Global services.

LEVEL 2 – Moderate / Slightly Difficult Questions

(Concept + Understanding + Internal Working)

1. How does traffic flow from browser to EC2?
2. Why can't a private subnet access the internet directly?
3. Difference between Security Group and NACL in real-world use?
4. What happens when Auto Scaling launches a new instance?
5. How does Load Balancer decide where to send traffic?
6. What is the difference between horizontal and vertical scaling?
7. How would you connect two VPCs?
8. What is the difference between IAM Role and IAM User?
9. How does DNS resolution work?
10. What is NAT Gateway and why is it used?
11. What are S3 storage classes and when would you use them?
12. How do you secure an EC2 instance?
13. What is Infrastructure as Code?
14. What is Docker and why is it used?
15. What is Kubernetes and why do we need it?

Topic: Networking & Traffic Flow

16. Explain step-by-step how traffic flows from the internet to a private EC2 instance behind a Load Balancer.
17. How would you allow a private EC2 instance to download updates securely?
18. How do route tables decide where traffic goes?
19. In what situation would you modify NACL instead of Security Group?
20. How would you connect an on-prem data centre to cloud securely?

Topic: Scaling & Availability

21. How does Auto Scaling detect unhealthy instances?
22. What happens if one Availability Zone fails in your architecture?
23. How would you ensure zero single point of failure?
24. How does a Load Balancer perform health checks?
25. When would vertical scaling be a bad idea?

Topic: Storage & Data

26. How do you decide between EBS and EFS for an application?
27. How would you design backup strategy for a production database?
28. How does object storage differ architecturally from block storage?
29. When would you move data to cheaper storage tiers?
30. How would you prevent accidental deletion of important data?

Topic: Security & IAM

31. How would you design least-privilege access for a team?
32. How do temporary credentials improve security?
33. How would you rotate secrets in a running application?
34. What security layers exist in a cloud architecture?
35. How would you isolate environments (dev, test, prod) securely?

Topic: DevOps & Containers (Optional but good to have basic understanding)

36. Why is containerization better than deploying directly on VMs?
37. What problem does Kubernetes solve that Docker alone doesn't?
38. How would you implement rolling deployment?
39. How does CI/CD reduce deployment risk?
40. How would you design logging and monitoring for microservices?

LEVEL 3 – Scenario-Based Questions (Most Important)

(Thinking + Troubleshooting + Architecture)

1. Production website is down. What are your first 5 steps?
2. EC2 is running but website not accessible. How do you debug?
3. CPU utilization suddenly spikes to 95%. What will you do?
4. Traffic increases 5x unexpectedly. What happens in your architecture?
5. How would you design a highly available web application?
6. How would you migrate an on-prem application to cloud?
7. Database is slow. How do you identify the bottleneck?
8. Users complain about slow performance. What will you check?

9. Kubernetes pod is in CrashLoopBackOff. Steps to debug?
10. How do you implement zero downtime deployment?
11. How would you reduce cloud cost in a running project?
12. Private EC2 cannot connect to RDS. How will you troubleshoot?
13. How do you design disaster recovery for production?
14. A new deployment caused errors in production. What will you do?
15. How would you secure a fintech/payment application architecture?

Topic: Networking & Connectivity Issues

16. A private EC2 cannot access the internet. How will you debug?
17. Users cannot access your public website. What layers will you check?
18. Traffic is reaching Load Balancer but not EC2. What could be wrong?
19. EC2 can ping internally but cannot connect to database. Why?
20. You connected two VPCs but instances cannot communicate. What will you verify?
21. Route tables look correct but traffic still not flowing. What next?

Topic: Scaling & Performance

22. Application slows down during peak hours. How will you investigate?
23. Auto Scaling is configured but instances are not scaling. Why?
24. CPU is low but response time is high. What might be the issue?
25. One Availability Zone goes down. What should happen in a properly designed system?
26. Users from another country complain about latency. How do you reduce it?

Topic: Architecture & Design

27. Design a highly available 3-tier architecture.
28. How would you eliminate single points of failure in a setup?
29. How would you design a system that can handle sudden traffic spikes?
30. How would you design backup and restore strategy for critical data?
31. How would you ensure minimal downtime during maintenance?
32. How would you design secure access for internal employees?

Topic: Security & IAM

33. A developer accidentally made an S3 bucket public. What will you do?
34. How would you implement least privilege access across teams?
35. An EC2 was compromised. What immediate steps will you take?
36. How would you securely allow application to access database credentials?
37. How would you audit who accessed sensitive resources?

Topic: Storage & Database

- 38. Database storage is full. What are your options?
- 39. How would you migrate data from one region to another safely?
- 40. How would you handle backup failure alerts?

Topic: Monitoring & Troubleshooting

- 41. Application is throwing random 500 errors. How do you approach it?
- 42. Logs show high memory usage but CPU is fine. What does that indicate?
- 43. Users report intermittent failures but metrics look normal. What next?
- 44. How would you design centralized logging for cloud resources?
- 45. How do you ensure alerts are meaningful and not noisy?

“Refer to the other document for Answers”

Before checking the answers, Test yourself and try to answer these questions

*******(without cheating 🙅)*******

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